CHAPTER 1 : Mathematical Modeling and Problem Solving

Numerical methods are techniques by which mathematical problems are formulated using arithmetic operations.

Noncomputer methods :

1) Analytical (Exact) Methods : used for a limited number of problems

2) Graphical Solutions : e.g. plots. Limited to problems in 3D or fewer dimensions.

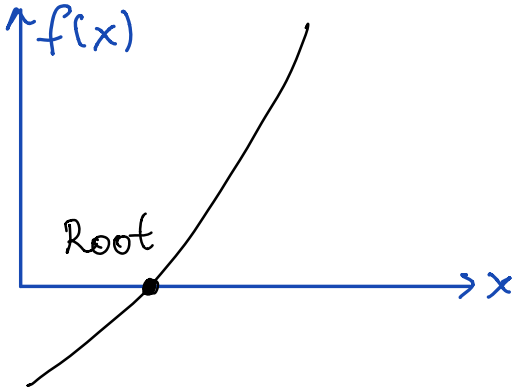
Why study Numerical Methods?

- NM are extremely powerful problem-solving tools.
- Use computer programs
- Learn how to program in MATLAB.
  - illustrate the power and limitations of computers
  - acknowledge and control errors of approximation.
- NM reinforce your understanding of mathematics by reducing higher mathematics to basic arithmetic operations.

Subject Areas to be Covered:

1) Roots of Equations

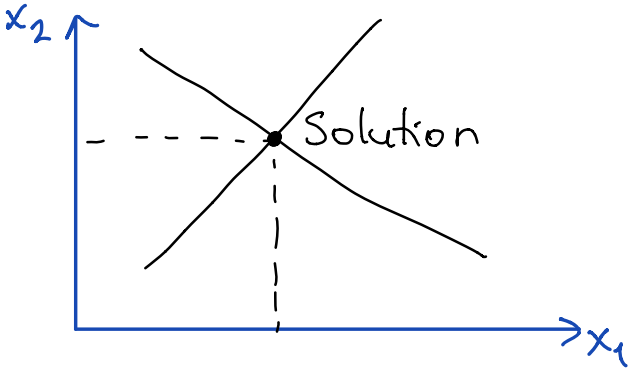
Solve $f(x) = 0$ for x



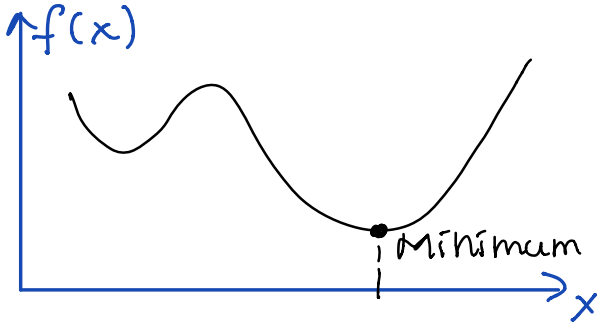
2) Systems of Linear Algebraic Equations

$$a_{11}x_1 + a_{12}x_2 = c_1$$
$$a_{21}x_1 + a_{22}x_2 = c_2$$

} Given a's, c's. Solve x's.



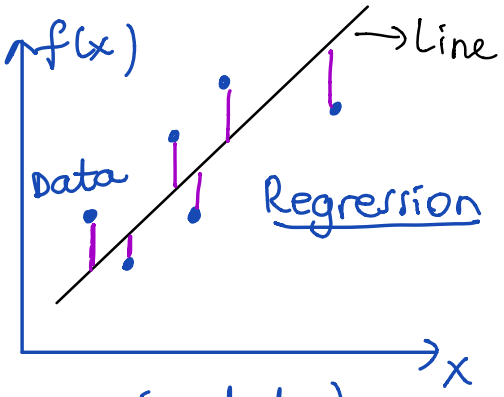
3) Optimization



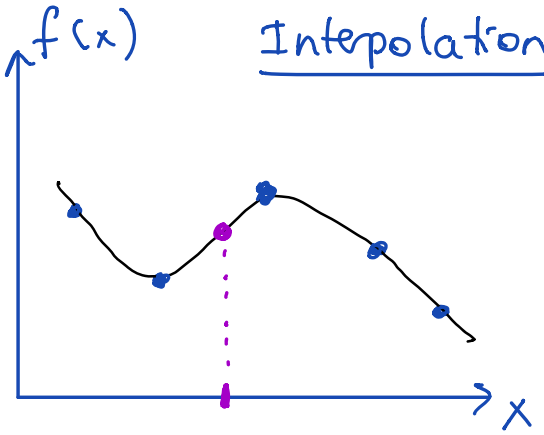
Determine x that gives optimum f(x)

- single and multi-variable unconstrained optimization.
- constrained optimization (linear programming)

4) Curve Fitting



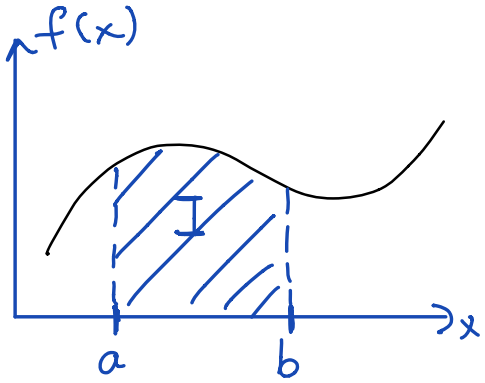
(error in data)



Interpolation

determine intermediate values for error-free data

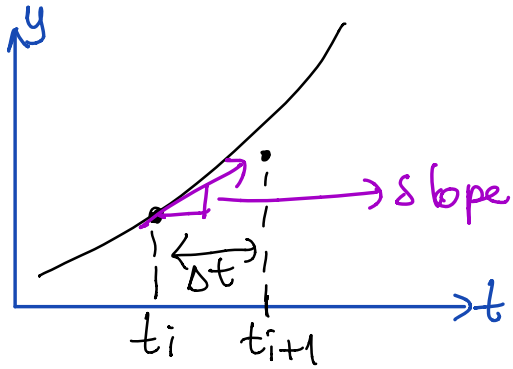
5) Integration



$$I = \int_a^b f(x)\,dx$$

Find the area under the curve

6) Ordinary Differential Equation



Given

$$\frac{dy}{dt} \cong \frac{\Delta y}{\Delta t} = f(t,y)$$

Solve for y as a function of t

$$y_{i+1} = y_i + f(t_i, y_i)\Delta t$$

Mathematical Model: a formulation or equation that expresses essential features of a physical system or process in mathematical terms.

$$\text{Dependent variable} = f\left(\begin{array}{l}\text{independent variables,}\\ \text{parameters,}\\ \text{forcing functions}\end{array}\right)$$

- dependent variable → reflects the state or behavior of the system
- independent variables → generally time or space
- forcing functions → external influences

EX Newton's second law of motion.

$F = ma$

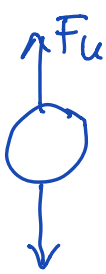
- a → acceleration (m/s^2)
- m → mass (kg)
- F → net force acting on a body (N, $kg\,m/s^2$)

$a = \frac{F}{m}$

- a → dependent variable reflect system's behavior
- F → forcing function
- m → parameter
- ⊛ No independent variable

⊛ Use Newton's second law to determine the terminal velocity of a free-falling body.

$$\frac{dv}{dt} = \frac{F}{m}$$, v: velocity (m/s)



$F_D = mg$, g: acceleration due to gravity ($9.81\ m/s^2$)

$F_U = -cv$, c: drag coefficient (kg/s)
(related to shape or surface roughness affect air resistance)

Combining

$$\frac{dv}{dt} = \frac{mg - cv}{m} = g - \frac{c}{m}v$$

$$\boxed{\frac{dv}{dt} = g - \frac{c}{m}v} \quad (1)$$ a differential equation

Analytical solution:

$$\boxed{v(t) = \frac{gm}{c}\left(1 - e^{-(c/m)t}\right)} \quad (2)$$ (exact solution)

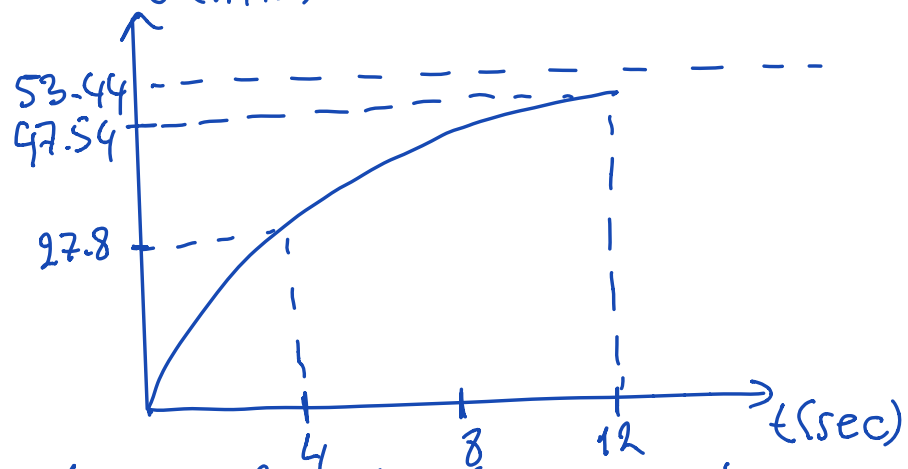
dependent var. → independent variable

Parachutist: mass = 68.1 kg
$c = 12.5\ kg/sec$
$g = 9.81\ m/s^2$
} inset in (2)

$$v(t) = \frac{(9.81)(68.1)}{(12.5)}\left(1 - e^{-(12.5/68.1)t}\right)$$

$$\boxed{v(t) = 53.44\left(1 - e^{-0.183t}\right)}$$ Analytical Solution

t(sec)	v(m/s)
0	0
2	16.42
4	27.80
⋮	⋮
10	44.92
12	47.54
∞	53.44 → terminal velocity is constant (eventually $F_D = F_u$ balanced with air resistance. Net force = 0)



Numerical Method: Reformulate the problem so that it can be solved by arithmetic operations.

For Newton's second law:

$$\frac{dv}{dt} \cong \frac{\Delta v}{\Delta t} = \frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i} \qquad (3)$$

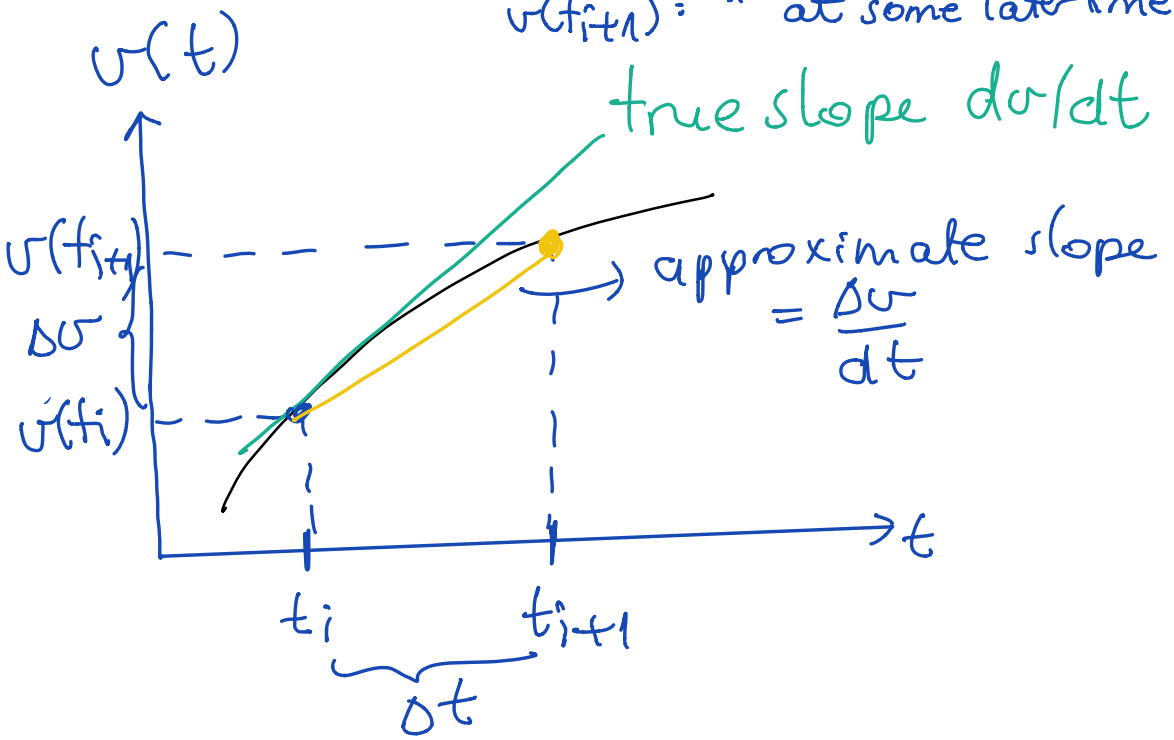
$$\frac{dv}{dt} = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t}$$

Δv: difference in velocity

Δt: difference in time

$v(t_i)$: velocity at initial time t_i

$v(t_{i+1})$: " at some later time t_{i+1}



(3): finite divided difference approximation of the derivative at time t_i

Substitute (3) into (1):

$$\frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i} = g - \frac{c}{m} v(t_i)$$

slope of yellow line

Numerical Solution

$$v(t_{i+1}) = v(t_i) + \left[g - \frac{c}{m} v(t_i)\right](t_{i+1} - t_i) \quad (4)$$

$v(t_{i+1})$: new value; $v(t_i)$: old value; $\left[g - \frac{c}{m} v(t_i)\right]$: slope; $(t_{i+1} - t_i)$: step size

Falling parachutist : $m = 68.1\ kg$

$c = 12.5\ kg/sec$

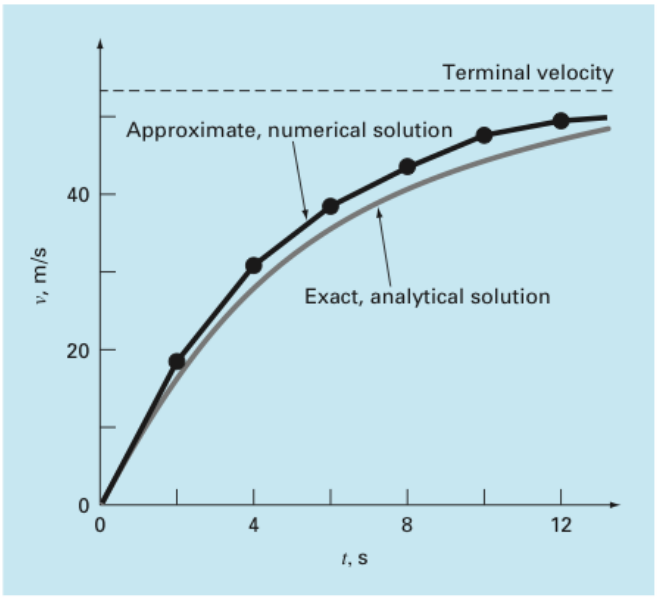
step size = $\Delta t = 2\ sec$

at $t_i = 0 \Rightarrow v = 0$

at $t_{i+1} = 2s \Rightarrow v = 0 + \left[9.81 - \frac{12.5}{68.1} \cdot 0\right] \cdot 2$

$= 19.62\ m/s$

t	v
0	0
2	19.62
4	32.04
⋮	⋮
10	48.02
12	50.01
∞	53.44



Comparison of analytical and numerical solution

* Can minimize differences by using a smaller step size.
* There's a tradeoff btwn accuracy and computational effort.